

Xiaoxiao YU

Agent Engineer | Full-Stack Engineer

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Full-stack agent engineer who ships production AI products end to end, from React/TypeScript frontends and Python/Node.js backends to the agent runtime underneath. At Viciking (full-time since July 2026), built the agent products that **doubled video-production efficiency**. Before: a browser-agent startup in Ant Group's MaShang AI Accelerator, zkWasm apps with thousands of users, and React Native apps.

Experience

Viciking (Beijing Weixi Network Technology)

Jul 2026 – Present

Agent Engineer, full-time | *sole engineer, frontend to agent runtime, on four internal AI production products*

- Versus manual editing, **production efficiency doubled**, with **~90% of automated outputs needing no manual re-editing**; dozens of internal users across multiple business lines.
- **WeiChuang Drama Studio** (React 19, FastAPI, Temporal, PostgreSQL): a 1-minute episode goes **from script to finished video in 15–30 min**, with episodes in parallel; **~90% need no manual re-editing**, the rest only light edits. If a model call fails mid-batch, only the failed shots rerun, so finished shots are never regenerated or paid for twice. Bad generations fall back to another model or get their voice repaired, and every model call is logged so a slow or stuck job can be traced to the exact call. ~5,000 tests.
- **Weixi Studio** (TypeScript, React, Remotion, PostgreSQL, MCP): turned OpenChatCut, a local-first single-user agentic video editor, into a **multi-tenant SaaS used across business lines**. Agent tasks survive worker crashes and resume where they stopped; a paid or irreversible tool call is never retried blindly, and if its outcome is unclear a person decides. Exposed the editor over MCP so outside agents can edit the same timeline as people without overwriting their changes.
- **Screenplay Agent Studio** (FastAPI, Pi agent SDK, OpenTelemetry): built an eval comparing one lead agent with a lead plus specialists; the specialists cost **3.7× the tokens for no quality gain**, so each of the 6 writing stages runs a single lead. Drafts live in files rather than chat history, so long sessions can be compressed without losing work; writers approve every edit; alerts fire when agent runs start failing or a decision waits too long.
- **Creative Engine** (TanStack Start, pgvector): a writing agent that drafts from analysed reference videos and keeps what it observed apart from what it inferred; ran 14 controlled experiments with an LLM judge and removed every prompt technique that did not measurably help.
- **Production tooling**: an unattended uploader took over submitting drama assets to the ad platform, work previously done by hand, with no duplicate submissions; a GPU redaction pipeline (fine-tuned RF-DETR) masks sensitive objects at **2× real time** with ~80% accuracy, still improving.

Delphinus Lab

Nov 2024 – Jul 2026

Developer Relations Engineer → Ecosystem Director (Jun 2025) | *Remote, concurrent with MPhil*

- Rebuilt the zkWasm browser SDK from scratch so apps use one wallet connection across both chain layers; wrote the CLI that scaffolds new apps; maintained the TypeScript services that settle deposits and withdrawals.
- Solo-built staking, launchpad, prediction-market and social-binder apps, plus the zkWasm Hub site and developer docs; ran DevOps for 8+ ecosystem apps, which reached **thousands of users** and **multi-million-USD TVL**.
- Led the team that built a hosted Kubernetes platform where developers deploy straight from GitHub and watch build status and logs live; it hosts **10+ projects**.

Earlier roles (2021–2024, alongside undergraduate studies)

- **Rosen** — Founding Full-Stack Engineer (2022–23): React / React Native apps for iOS, Android and Web; backed by PayPal and Circle. **2PM.Network** — Co-Founder & CEO (2024). **HashPunk** — Product Manager (2021–22), raised US\$1M. **1CIRCLE** — Full-Stack Engineer (2021–22). **Conflux** — Intern (2021).

Selected Projects

Resona — Browser Agents for Product Demos

May – Jun 2026

Own startup, sole engineer | *Ant Group MaShang AI Accelerator; trial customers* | *Stagehand, Playwright, AgentBay, BullMQ*

- Give it a product URL and it returns a demo video: browser agents map the product, explore each section in parallel in cloud sandboxes, track what they have covered and go back for anything missed. Any action that changes data needs a check first, and payments are blocked outright.
- When an agent hits a login page, the user signs in once in the same browser and the agent carries on from there. Explore, record and render run as separate job queues; ~500 tests.
- Render agents (GSAP, Remotion) cut the recordings into the finished demo; each step picks its own model from OpenAI, Anthropic or Gemini.

- Designed 17 role agents and 25 skills that take a novel through ingest, analysis, episode planning, writing, review and storyboard / video prompts. A review director gates every episode, and failures loop back for rewrite and re-review.
- Progress is saved to files, so a run can stop and resume at any episode without the context window filling up; hybrid ChromaDB + TF-IDF search pulls similar reference scripts to ground the writing.

ShotLoom — AI Film Shot-Breakdown Workspace

Open source, sole author | FastAPI, Redis/RQ, React, TypeScript, Qwen vision + ASR

- Built a shot-by-shot film breakdown: scene detection and clip cutting, then Qwen vision analysis of shot size, camera movement, lighting and narrative function, aligned with ASR transcripts and a whole-film continuity report.
- Long films are split into overlapping chunks so no scene is lost at a boundary; progress streams live and survives a page refresh; Excel/PDF export; 107 end-to-end tests.

RingChat — Agent-Native Messaging

Jan – Mar 2026

Team lead, client engineer | Kotlin / Jetpack Compose, Web, iOS | OpenClaw (self-hosted agent gateway)

- Led a **6-person team** building a cross-platform IM in which every user gets a personal AI agent, and worked mainly on the clients (now continued by another team as Gold House). On Android, built the chat UI, optimistic media upload, contact and group search, unread badges and multi-device sync.

Skills

Agents	Context engineering, tool and MCP design, multi-agent orchestration, browser agents (Stagehand, Playwright, AgentBay), human-in-the-loop, evals and LLM-as-judge
Frontend & mobile	TypeScript, React 19, TanStack (Start / Query), Next.js, Vite, Remotion; React Native, Kotlin / Jetpack Compose
Backend	Python (FastAPI), Node.js; PostgreSQL / pgvector, Redis, Temporal, BullMQ, SSE, WebSocket
Infra & models	Docker, Kubernetes, CI/CD, OpenTelemetry; OpenAI, Anthropic, Gemini, Qwen, Doubao/Seedance via Vercel AI SDK and gateways
Other	Solidity, Rust, C++ (ESP32), PyTorch, scikit-learn

Education

The Hong Kong University of Science and Technology (Guangzhou)

Sep 2024 – Oct 2026

Red Bird MPhil in Data Science and Analytics | graduated Oct 2026 | thesis: local-first retrieval over 1.36M Mastodon posts, 79.3% accuracy at under 100 ms per query

The University of Edinburgh

Sep 2020 – Jun 2024

MA in Cognitive Science, First Class Honours | Exchange: Fudan University, Computer Science (2022–23)

Publication

Xiaoxiao YU, L. Du, C. Zeng, L. Deng, H. Luo, X. Gao, J. Tang. *SocialAttributionQA: A Benchmark for Attribution-Evidence-Aware Social Web Question Answering*. **WISE 2026** (CCF B), accepted; Springer LNCS. First author: built a 1,200-question benchmark over 725k Mastodon posts and the ATLAS graph-plus-retrieval solver (**95.3%** accuracy), with ablations and error analysis.

Selected Hackathons & Honours

33 hackathon and competition awards (12 first place or champion); AI and full-stack highlights:

Ava Box — Voice-First Trading Agent

AVE CLAW Hackathon 2026 — Champion

Sole developer | ESP32-S3, C++, Python, LVGL

- Built a handheld voice agent that routes speech to trading tools based on the current screen; every order needs a button confirmation, and the LVGL UI is shared between firmware and a desktop simulator. Extracted into DeviceKit, an open framework where the chain, speech and language models can be swapped, with over-the-air updates.

OutcomeX — AI Delivery & Settlement

HashKey Chain Horizon 2026 — 2nd, DeFi Track

Sole developer | FastAPI, AgentSkillOS, Solidity

- Built a marketplace where buyers pay for an AI deliverable: an agent runtime (AgentSkillOS) does the work with live progress, and payment is held in escrow until the buyer accepts, with refunds and revenue sharing in 7 Foundry contracts.

2025	Zama Bounty, Universal FHEVM SDK — Winner	2024	BeWater AI Crypto — 1st, Security & Privacy
2025	ICBC Cup FinTech, Guangdong — First Prize	2024	ETH Beijing — 1st, Scroll Track
2024	ETH Shanghai — Overall Champion	2021	PaddlePaddle Anti-Fraud ML — 1st (solo)

Founded LambdaSoc (Edinburgh) and Lambda Web3 (**1,000+ members**); **7,000+** Xiaohongshu followers for engineering content. Languages: Chinese (native), English (professional).